

Week .3

Course Name : Problem Solving and Testing

Course Code : 10211CS227

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1.Java Date and Time

2.Number of Days Between Two Dates

3.Day of the Year

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5.Java Priority Queue

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1.Java Date and Time

PROGRAM:

```
import java.util.*;

public class Main {

    public static String findDay(int month, int day, int year) {
        Calendar cal = Calendar.getInstance();

        cal.set(year, month - 1, day);

        String[] days = {
            "SUNDAY",
            "MONDAY",
            "TUESDAY",
            "WEDNESDAY",
            "THURSDAY",
            "FRIDAY",
            "SATURDAY"
        };

        return days[cal.get(Calendar.DAY_OF_WEEK) - 1];
    }
}
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
  
    int month = sc.nextInt();  
    int day = sc.nextInt();  
    int year = sc.nextInt();  
  
    System.out.println(findDay(month, day, year));  
  
    sc.close();  
}  
}
```

OUTPUT:



```
21 05 2006  
WEDNESDAY  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

2.Number of Days Between Two Dates

PROGRAM:

```
import java.time.LocalDate;

import java.time.temporal.ChronoUnit;

class Solution {

    public int daysBetweenDates(String date1, String date2) {

        LocalDate d1 = LocalDate.parse(date1);

        LocalDate d2 = LocalDate.parse(date2);

        return (int) Math.abs(ChronoUnit.DAYS.between(d1, d2));

    }

}
```

OUTPUT:

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date1 =
"2019-06-29"

date2 =
"2019-06-30"

Output

1

Expected

1

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date1 =
"2020-01-15"

date2 =
"2019-12-31"

Output

15

Expected

15

3.Day of the Year

PROGRAM:

```
import java.time.LocalDate;

class Solution {
    public int dayOfYear(String date) {
        LocalDate d = LocalDate.parse(date);
        return d.getDayOfYear();
    }
}
```

OUTPUT:

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date =
"2019-01-09"

Output

9

Expected

9

Accepted Runtime: 4 ms

✓ Case 1 ✓ Case 2

Input

date =
"2019-02-10"

Output

41

Expected

41

4.Day of the Week

PROGRAM:

```
import java.time.LocalDate;

class Solution {

    public String dayOfTheWeek(int day, int month, int year) {

        LocalDate date = LocalDate.of(year, month, day);

        String dayName = date.getDayOfWeek().toString();

        return dayName.charAt(0) + dayName.substring(1).toLowerCase();

    }

}
```

OUTPUT:

Accepted Runtime: 5 ms

Case 1 Case 2 Case 3

Input

day =
31

month =
8

year =
2019

Output

"Saturday"

Expected

"Saturday"

Accepted Runtime: 5 ms

Case 1 Case 2 Case 3

Input

day =
18

month =
7

year =
1999

Output

"Sunday"

Expected

"Sunday"

Accepted Runtime: 5 ms

Case 1 Case 2 Case 3

Input

day =
15

month =
8

year =
1993

Output

"Sunday"

Expected

"Sunday"

5.Java Priority Queue

PROGRAM:

```
import java.util.*;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }
}
```

```
public double getCGPA() {  
    return cgpa;  
}  
}
```

```
class Priorities {
```

```
public List<Student> getStudents(List<String> events) {
```

```
    PriorityQueue<Student> pq = new PriorityQueue<>(  
        (a, b) -> {
```

```
            if (a.getCGPA() != b.getCGPA()) {
```

```
                return Double.compare(b.getCGPA(), a.getCGPA());
```

```
            }  
        });
```

```
    }
```

```
    int nameCompare = a.getName().compareTo(b.getName());
```

```
    if (nameCompare != 0) {
```

```
        return nameCompare;
```

```
    }
```

```
    return Integer.compare(a.getID(), b.getID());
```

```
    }
```

```
};
```



```
for (String event : events) {

    if (event.startsWith("ENTER")) {

        String[] parts = event.split(" ");

        String name = parts[1];

        double cgpa = Double.parseDouble(parts[2]);

        int id = Integer.parseInt(parts[3]);

        pq.add(new Student(id, name, cgpa));

    } else if (event.equals("SERVED")) {

        if (!pq.isEmpty()) {

            pq.poll();

        }

    }

}

List<Student> result = new ArrayList<>();

while (!pq.isEmpty()) {

    result.add(pq.poll());

}

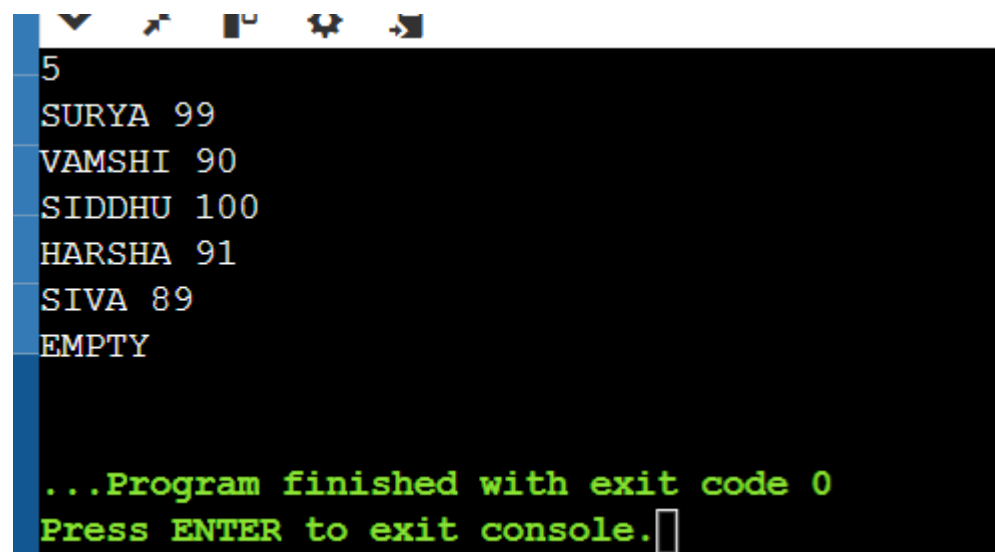
return result;
```

```
}  
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
        sc.nextLine();  
  
        List<String> events = new ArrayList<>();  
  
        for (int i = 0; i < n; i++) {  
            events.add(sc.nextLine());  
        }  
  
        Priorities priorities = new Priorities();  
        List<Student> students = priorities.getStudents(events);  
  
        if (students.isEmpty()) {  
            System.out.println("EMPTY");  
        } else {  
            for (Student student : students) {  
                System.out.println(student.getName());  
            }  
        }  
    }  
}
```

```
    }  
}  
  
    sc.close();  
}  
}
```

OUTPUT:

A screenshot of a terminal window with a black background and a blue vertical bar on the left. The window title bar shows standard icons. The output text is as follows:

```
5  
SURYA 99  
VAMSHI 90  
SIDDHU 100  
HARSHA 91  
SIVA 89  
EMPTY  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

6.Java ArrayList

PROGRAM:

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<ArrayList<Integer>> list = new ArrayList<>();

        // Read the lines
        for (int i = 0; i < n; i++) {

            int size = sc.nextInt();

            ArrayList<Integer> row = new ArrayList<>();

            for (int j = 0; j < size; j++) {

                row.add(sc.nextInt());

            }

            list.add(row);

        }

    }

}
```

```

    }

    // Number of queries
    int q = sc.nextInt();

    for (int i = 0; i < q; i++) {
        int x = sc.nextInt();
        int y = sc.nextInt();

        // x and y are 1-based
        if (x >= 1 && x <= list.size() &&
            y >= 1 && y <= list.get(x - 1).size()) {

            System.out.println(list.get(x - 1).get(y - 1));

        } else {
            System.out.println("ERROR!");
        }
    }

    sc.close();
}
}

```

OUTPUT:

```
Console I/O AI Agent
5
5 41 77 74 22 44
1 12
4 37 34 36 52
0
3 20 22 33
5
1 3
74
3 4
52
```

7.Largest Number

PROGRAM :

```
import java.util.*;

class Solution {
    public String largestNumber(int[] nums) {

        String[] arr = new String[nums.length];

        for (int i = 0; i < nums.length; i++) {
            arr[i] = String.valueOf(nums[i]);
        }

        Arrays.sort(arr, (a, b) -> (b + a).compareTo(a + b));

        if (arr[0].equals("0")) {
            return "0";
        }

        StringBuilder result = new StringBuilder();

        for (String s : arr) {
            result.append(s);
        }
    }
}
```

```
    }  
  
    return result.toString();  
}  
}
```

OUTPUT :

Accepted Runtime: 1 ms	Accepted Runtime: 1 ms
☒ Case 1	☒ Case 1
☒ Case 2	☒ Case 2
Input	Input
nums = [10,2]	nums = [3,30,34,5,9]
Output	Output
"210"	"9534330"
Expected	Expected
"210"	"9534330"

8.Java Comparator

PROGRAM :

```
import java.util.*;

class Player {
    String name;
    int score;

    Player(String name, int score) {
        this.name = name;
        this.score = score;
    }
}

class Checker implements Comparator<Player> {

    @Override
    public int compare(Player a, Player b) {

        // Sort by decreasing score
        if (a.score != b.score) {
            return b.score - a.score;
        }
    }
}
```

```
        // If scores are same, sort alphabetically
        return a.name.compareTo(b.name);
    }
}
```

```
public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        Player[] players = new Player[n];

        for (int i = 0; i < n; i++) {
            String name = sc.next();
            int score = sc.nextInt();

            players[i] = new Player(name, score);
        }

        Arrays.sort(players, new Checker());

        for (Player p : players) {
            System.out.println(p.name + " " + p.score);
        }
    }
}
```

```
}  
  
    sc.close();  
}  
}
```

OUTPUT :

☰ Console

📄 I/O

🌟 AI Agent

4
surya 500
vamshi 600
siva 700
siddhu 900
siddhu 900
siva 700
vamshi 600
surya 500

9.Sort the People

PROGRAM :

```
import java.util.*;

class Solution {
    public String[] sortPeople(String[] names, int[] heights) {

        Integer[] index = new Integer[names.length];

        for (int i = 0; i < names.length; i++) {
            index[i] = i;
        }

        Arrays.sort(index, (a, b) -> heights[b] - heights[a]);

        String[] result = new String[names.length];

        for (int i = 0; i < names.length; i++) {
            result[i] = names[index[i]];
        }

        return result;
    }
}
```

```
}
```

OUTPUT :

Accepted Runtime: 1 ms

✓ Case 1

✓ Case 2

Input

names =
["Mary", "John", "Emma"]

heights =
[180, 165, 170]

Output

["Mary", "Emma", "John"]

Expected

["Mary", "Emma", "John"]

Accepted Runtime: 1 ms

✓ Case 1

✓ Case 2

Input

names =
["Alice", "Bob", "Bob"]

heights =
[155, 185, 150]

Output

["Bob", "Alice", "Bob"]

Expected

["Bob", "Alice", "Bob"]